Stat 135 Content Rough Breakdown

Syllabus	Topics

Unit 1: Parametric inference
Weeks 2-5
Chapter 8 of Rice
	- What is a **parametric model**? (8.1-8.3)
	- What methods are available to estimate parameters from data?
	  - **MLE** (8.5), **MOM** (8.4)
	- How to quantify uncertainty of our estimates?
	  - **Standard error** computation
	  - Construction of confidence intervals (8.5): **bootstrap** (Adam's notes), **asymptotic theory of MLE** + **Fisher information** (8.5)
	- Unbiased estimators satisfy **CRLB** for variance (8.7)
	- How to obtain **sufficient statistic**? I.e., how much can data be compressed without losing information? (8.8)
	- **Rao-Blackwell method** produces optimal estimators (8.8)
	- Difference between **minimal sufficiency** and sufficiency (Adam's notes)
	- **Delta method** for functions of MLE (4.6, Adam's notes)

Unit 2: Hypothesis testing (I) - one sample only Weeks 6, 9 Chapter 9 of Rice	- What are **null hypothesis** and **alternative hypothesis**? (9.2) - What are **simple hypothesis** and **composite hypothesis** (9.2) - What are **p values**, **significance levels** and **power**? (9.2) - What are **Type I** and **Type II errors**? (9.2) - **Neyman-Pearson lemma** implies maximum power of **LRT** (9.2) - **Uniformly most powerful** (test) means most powerful for every simple alternative (9.2) - **Duality of CI and hypothesis testing**, rejecting a particular hypothesis corresponds to checking if the CI contains a value (9.3) - **Generalized likelihood ratio (Λ) test** and convergence of **-2 log Λ** to chi-square distribution with **appropriate degrees of freedom** (9.4) *(Read Adam's notes to supplement Rice)*

Unit 3: Hypothesis testing (II) - comparing multiple samples
Weeks 7, 8, 11, 12
Chapters 11-13 of Rice
	(You should ideally be familiar with Unit 2, because Unit 3 builds on Unit 2 conceptually.)
	3.1 Comparing continuous / quantitative data sets
	- **Parametric approach**: samples are normally distributed and independent
	  - **Two-sample z-test** if variances are known (11.2)
	  - **Two-sample t-test** if variances are unknown
	  - If variances assumed equal, use **pooled variance** (11.2)
	  - If variances not assumed equal, compute the overall variance and perform **Welch's t-test** (11.2)
	  - Know the **degrees of freedom** for each scenario
	  - **Paired samples** can either reduce or inflate variance, depends on correlation of pairs (11.3)
	- **Non-parametric approach**: don't know distributions from which independent samples are obtained
	  - If samples are **paired** (e.g., before vs after or longitudinal

	studies), perform **Wilcoxon signed rank** test (11.3) - If samples are **unpaired**, perform **Mann-Whitney (aka Wilcoxon rank sum)** test (11.2) 3.2 Comparing categorical data sets - **Chi-square test** - **Goodness of fit** (Adam's notes) - **Homogeneity** (13.3) - **Independence** (13.4) - When to use which test? - Know **degrees of freedom** for each test 3.3 Comparing more than two samples (continuous) - Parametric approach: errors normally distributed and independent, equal variances - **ANOVA** has **F statistic** under the null (12.2) - Know **degrees of freedom** - Non-parametric approach: don't know distributions from which independent samples are obtained - **Kruskal-Wallis test** (12.2) - **Post-hoc tests**, i.e., what happens after your ANOVA or Kruskal-Wallis returns a significant p value - Control of **Family-Wise Error Rate** using **Bonferroni method** (12.2.2)
Unit 4: Linear models Weeks 13, 14 Chapter 14 (focus on 14.2 and 14.5) of Rice	- Assumptions of the linear model: **normality**, **equal variances** - X is given, **no need to assume X is normal**. Only **y is normally distributed**! - **Homoscedasticity** (due to equal variance assumption) vs **heteroscedasticity** (violation of assumption) - Goal is to **estimate the coefficients β from data** and to **potentially use the estimates to make predictions on new data** - Point estimate: **hat(β) = $(X^TX)^{-1}X^Ty$** (14.5) - Need X to be **full rank** (what happens if p>n?) - It is **unbiased** - It minimizes **mean square error** - Uncertainty of estimate: **Cov(hat(β)) = $\sigma^2(X^TX)^{-1}$**, but usually σ^2 not known, so estimated by **hat(σ^2) = RSS/(n-p)**, where **p is number of predictors being estimated** - What's the distribution of **normalized β_i's**? (t-distributed, because σ^2 is estimated) - Can be used to **construct confidence intervals** for β_i's - Can be used for **testing hypotheses** about β_i's
Unit 5: Bayesian statistics Weeks 15 Adam's lecture notes	*(This unit is thematically related to Unit 1 through the common goal of inference. It is thematically related to Unit 2 through hypothesis testing. Read Adam's notes.)* - Departure from frequentist framework in parametric inference: there is

	no "true" θ, but a random variable Θ with its distribution (called the **prior distribution**)
	- **Bayes' rule**: P(A|B) = P(A & B) / P(B)
	- How to **compute posterior distribution**
	  - **Posterior $\propto$ Likelihood x Prior**
	  - Inspect posterior and identify distribution (e.g., if it's of the form $\theta^k(1-\theta)^m$ for natural numbers k and m, with $0 \leq \theta \leq 1$, then it's beta(k+1,m+1) distributed)
	- Use posterior distribution to **compute posterior means and variances**
	- **Bayesian credible interval** vs frequentist confidence interval
	  - What are their interpretations?
	- **Bayesian hypothesis testing** (see examples and explanation here)

<u>Other topics</u>

Techniques

- when **transforming densities**, use the change of variables formula (aka Jacobian method)
- **Bootstrap**: both **parametric** and **non-parametric**, what are their differences? When to use either method?
- Be familiar with integration by parts and **differentiation + setting derivatives to 0**